Meeting 1015

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資料介紹

SECOM

- 樣本數 1567
- 變數 Time ／ x1 ／ x2 ／... ／ x590 ／ PassFail
  - Features: 591, Y: PassFail (binary classification)
  - Time: 2008-07-19 11:55:00
  - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
  - Missing values: 41951

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     2. KNN-imputation
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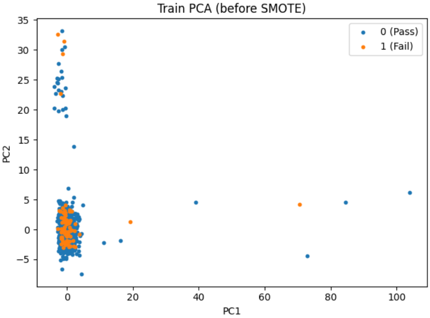
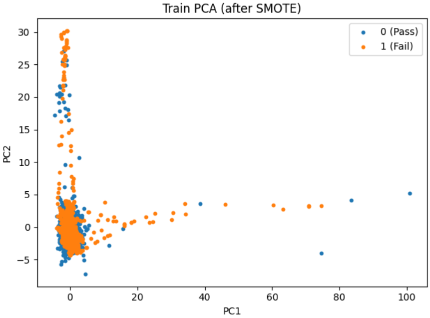
文獻回顧 I-方法 (SMOTE)



SMOTE before



SMOTE after

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     5. 0.8 training, 0.2 testing by decision tree
  2. 結果
     1. Best: ACC 86.94%, FAR 9.22%
     2. 反查 Fail 節點

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研究動機

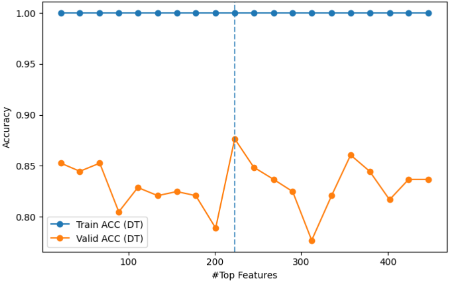
- K-folds cross validation
  - Split to training&testing first
  - Robust performance
- Metric
  - ACC, FAR, Sensitivity, Specificity
- $y = \hat{f}(x)$
  - Important features

研究結果（Decision Tree（Test）$k = 223$）

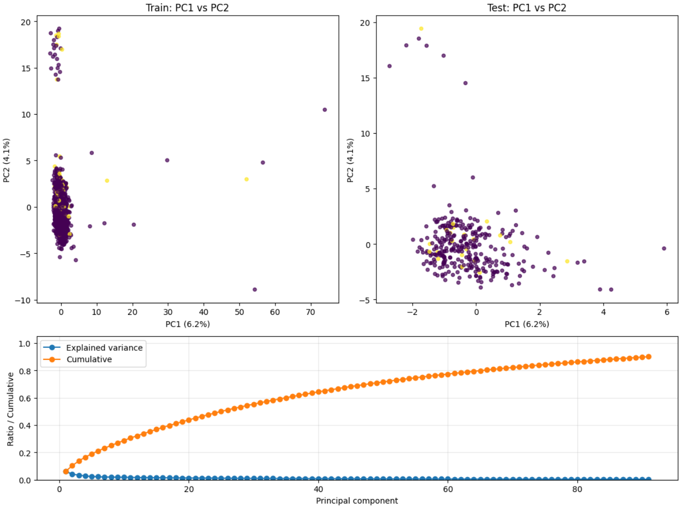
Accuracy	0.8662
FAR	0.0922
Sensitivity	0.2857
Specificity	0.9078
GM	0.5093

Confusion Matrix

	Pred 0	**Pred 1**
True 0	266	27
True 1	15	6



研究結果


Train: PC1 vs PC2
Test: PC1 vs PC2
PC2 (4.1%)
PC1 (6.2%)
Explained variance
Cumulative
Ratio / Cumulative
Principal component

參考資料

Li-Fu Lai (2022)。

使用 KNN 和決策樹方法對填補缺失值的製造數據進行失效預測和原因追蹤分析

The End

Questions & Comments